

Visualisation

Lecture 3

Encodings 1

Dr. Marisa Llorens

Food and gas dominate the price increases Americans have noticed.

Nearly 1,900 survey respondents listed specific items for which they've noticed rising prices in the past year. The numbers show how many people mentioned each item. Food items are highlighted.



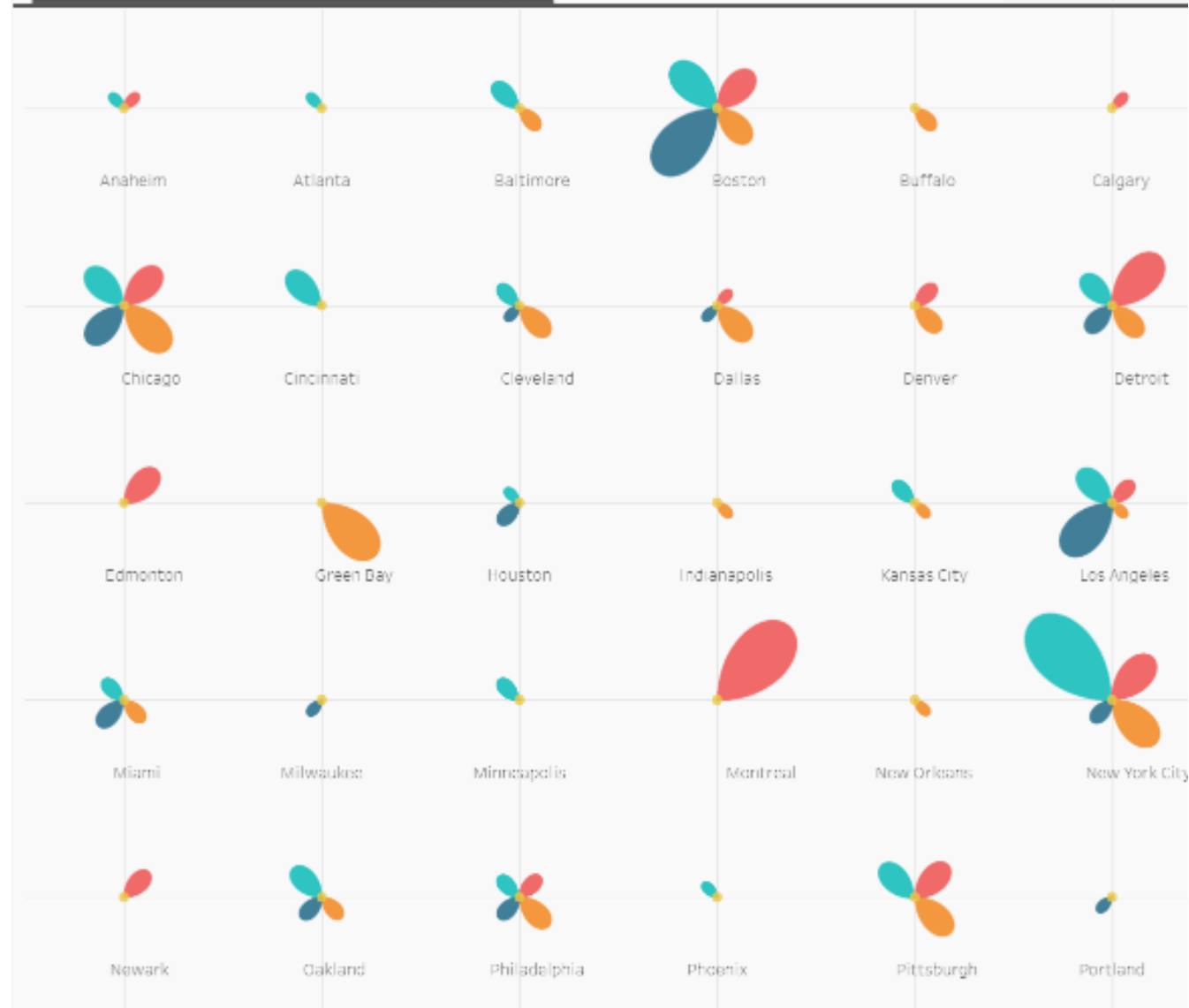
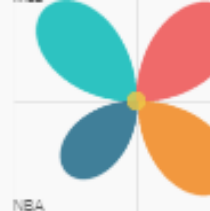
SPORTS CITY



Have you ever wondered which cities have won the most championships in the four major sports leagues? In this visualization, each petal of the flower chart represents one of the four major sports leagues. The size of the petal represents the number of championships in the respective league.

Click on the petals to the right to highlight the league in all cities; click on the center to see total championships.

Hover over each petal in the chart below to see the breakdown of championships by team through 2018.



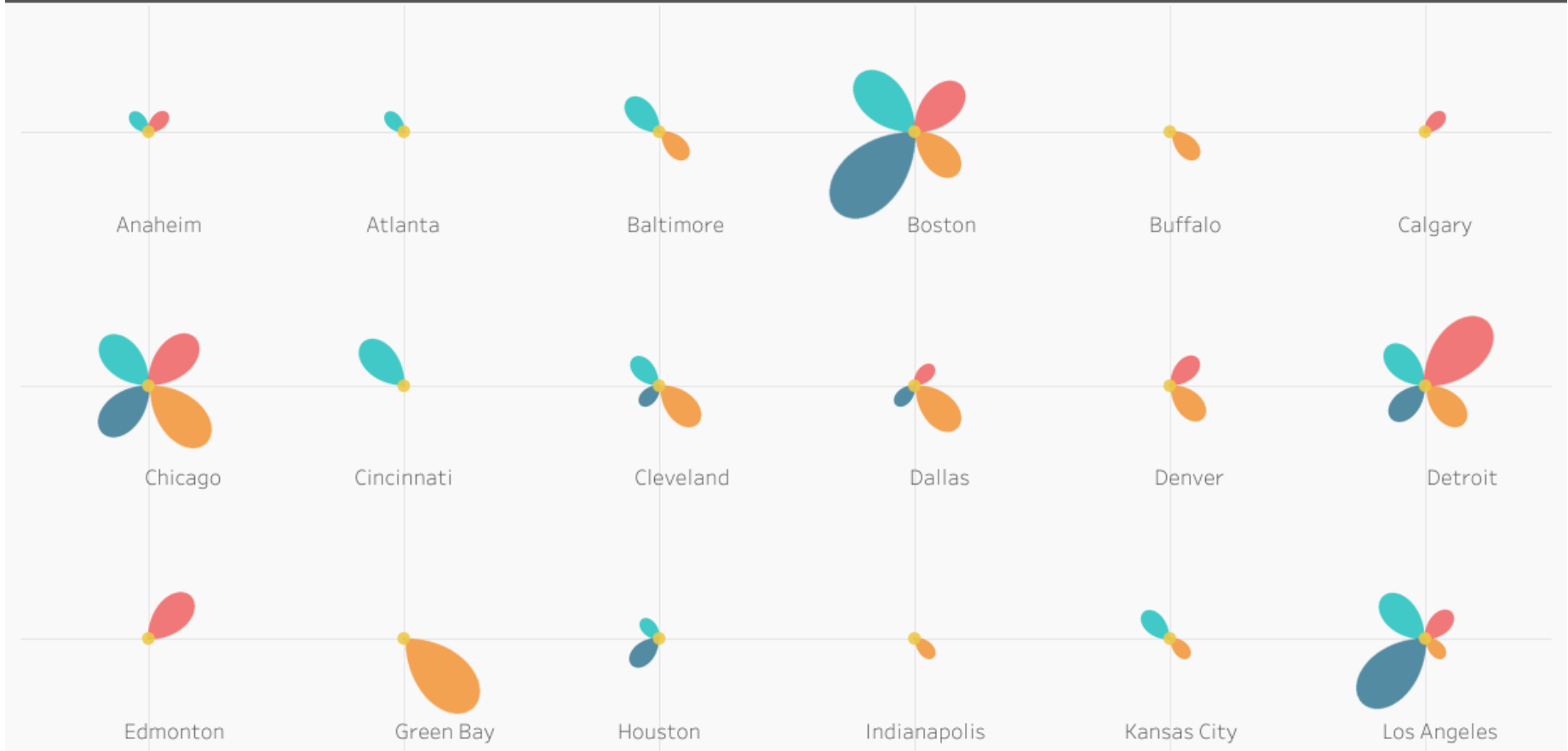
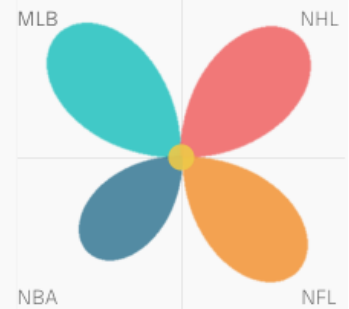
SPORTS CITY



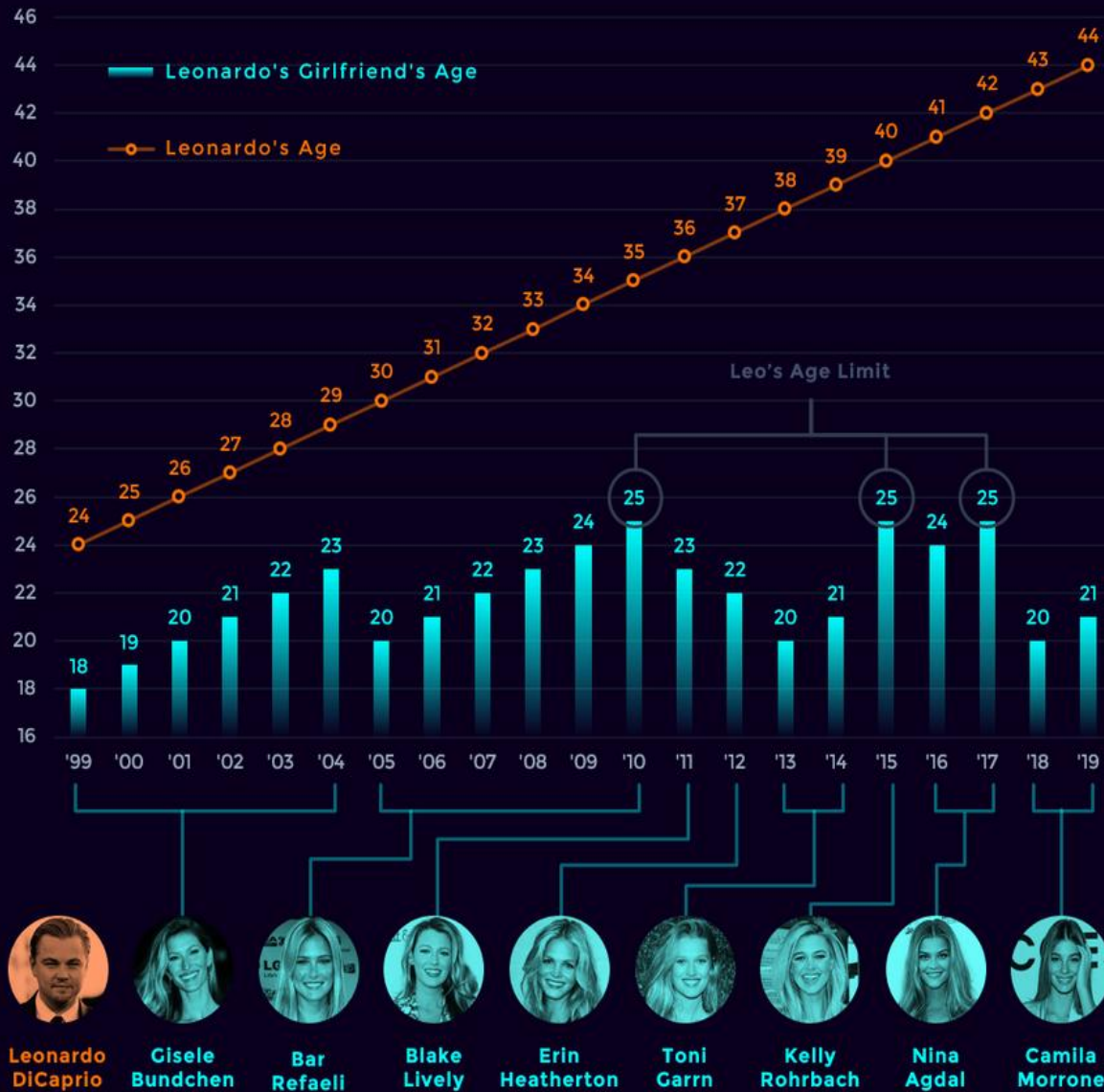
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Click on the petals to the right to highlight the league in all cities; click on the center to see total championships.

Hover over each petal in the chart below to see the breakdown ..



LEONARDO DICAPRIO REFUSES TO DATE A WOMAN OVER 25



Agenda

Data visualisations fundamentally map data dimensions to visual encodings

In today's lecture we will look at:

- The different encodings available and how we can appropriately use them

DIMENSIONS & ENCODINGS

Data Types Classification

- Categorical(qualitative): no inherent order:
city names, types of diseases, ...
- Ordinal(qualitative): ordered, but not at measurable intervals
first, second, third, ...
cold, warm, hot
Mon, Tue, Wed, Thu ...
- Quantitative: Integers or Reals
1, 2, 7
3.4, 78.6, -12.4

Data Representation

Data representation is the act of giving visual form to your data.

- ➡ Viewers decode the various shapes, sizes, positions and colours to form an understanding of the values presented.
- ⬅ Visualisers do the reverse through visual encodings, assigning visual properties to values.

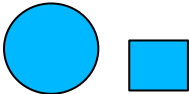
Data Representation

Visual encoding of data involves:

- Marks
- Attributes

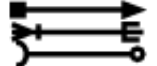
The objective of visual encoding is to find the right blend of marks and attributes that most effectively will portray the angle of analysis you wish to show your viewers.

Marks

Point		No variation
Line		One linear spatial dimension
Area		Two spatial dimensions
Form		Three spatial dimensions

Attributes

Position
Size
Colour hue
Colour Saturation
Colour Lightness
Symbol/Shape
Pattern
Angle/Slope
....and others...

Example	Encoding	Ordered
	position, placement	yes
1, 2, 3; A, B, C	text labels	optional alpha or num
	length	yes
	size, area	yes
	angle	yes
	pattern density	yes
	weight, boldness	yes
	saturation, brightness	yes
	color	no
	shape, icon	no
	pattern texture	no
	enclosure, connection	no
	line pattern	no
	line endings	no
	line weight	yes

MOST EFFECTIVE

Position on a scale



Length (1D size)



Direction



Angle



Area (2D size)



Volume (3D size)



Curvature



Luminance/shading

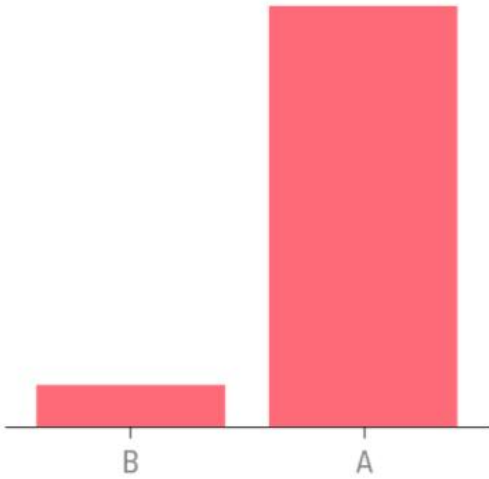


Color saturation



LEAST EFFECTIVE

Visual challenge



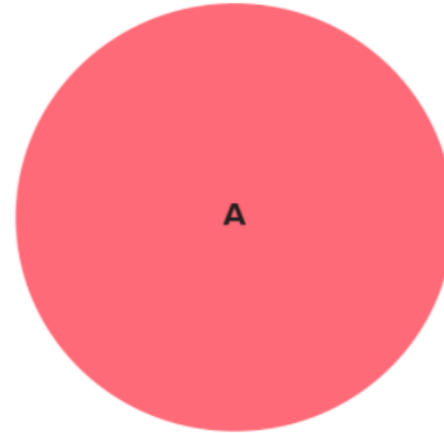
The value of A is how many times as large as the value of B?

9

8

10

5



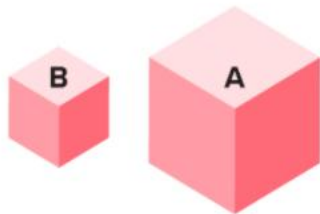
The area of circle A is how many times as large as the area of circle B?

4

7

10

14



The volume of cube A is how many times as large as the volume of cube B?

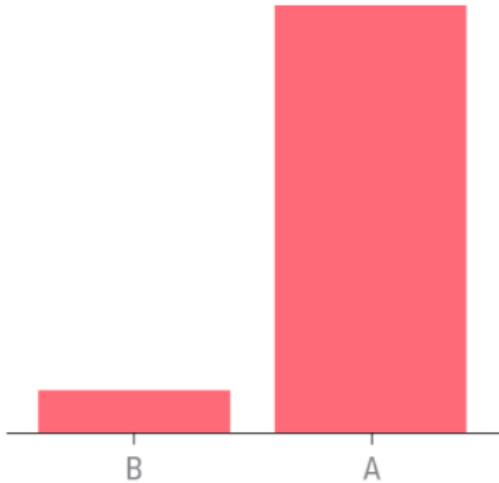
3

5

7

8

Visual challenge



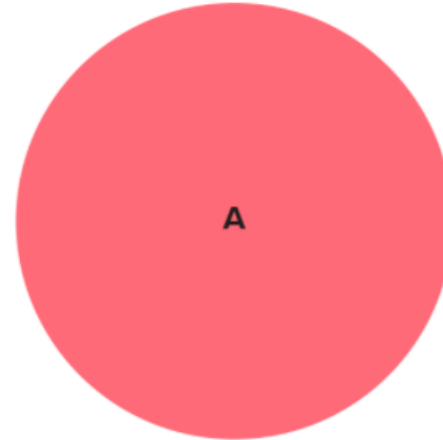
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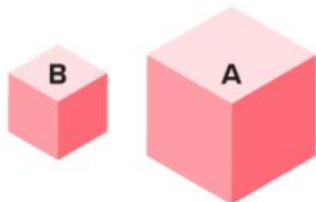
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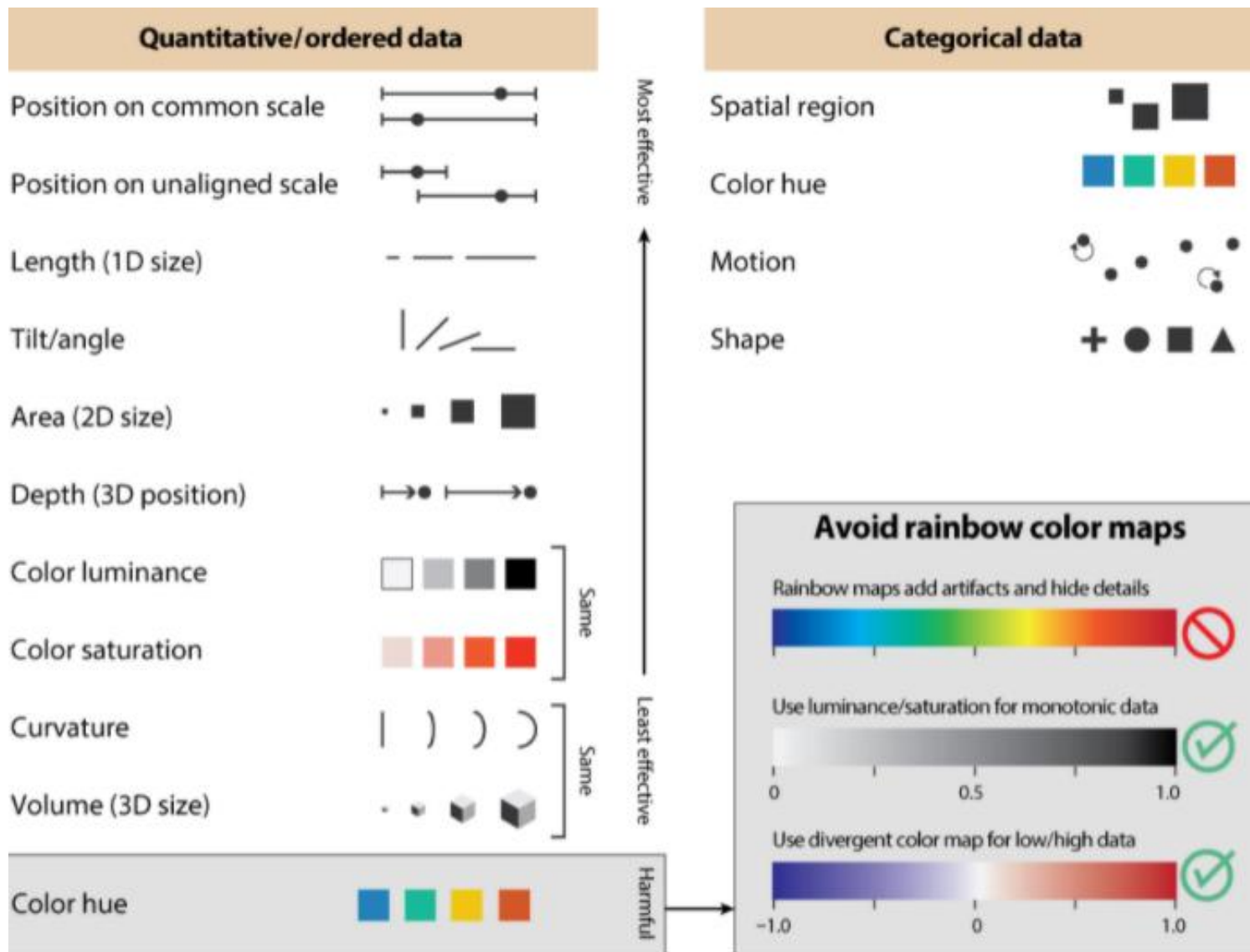
7

8

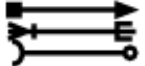
Interpretations of Visual Encodings

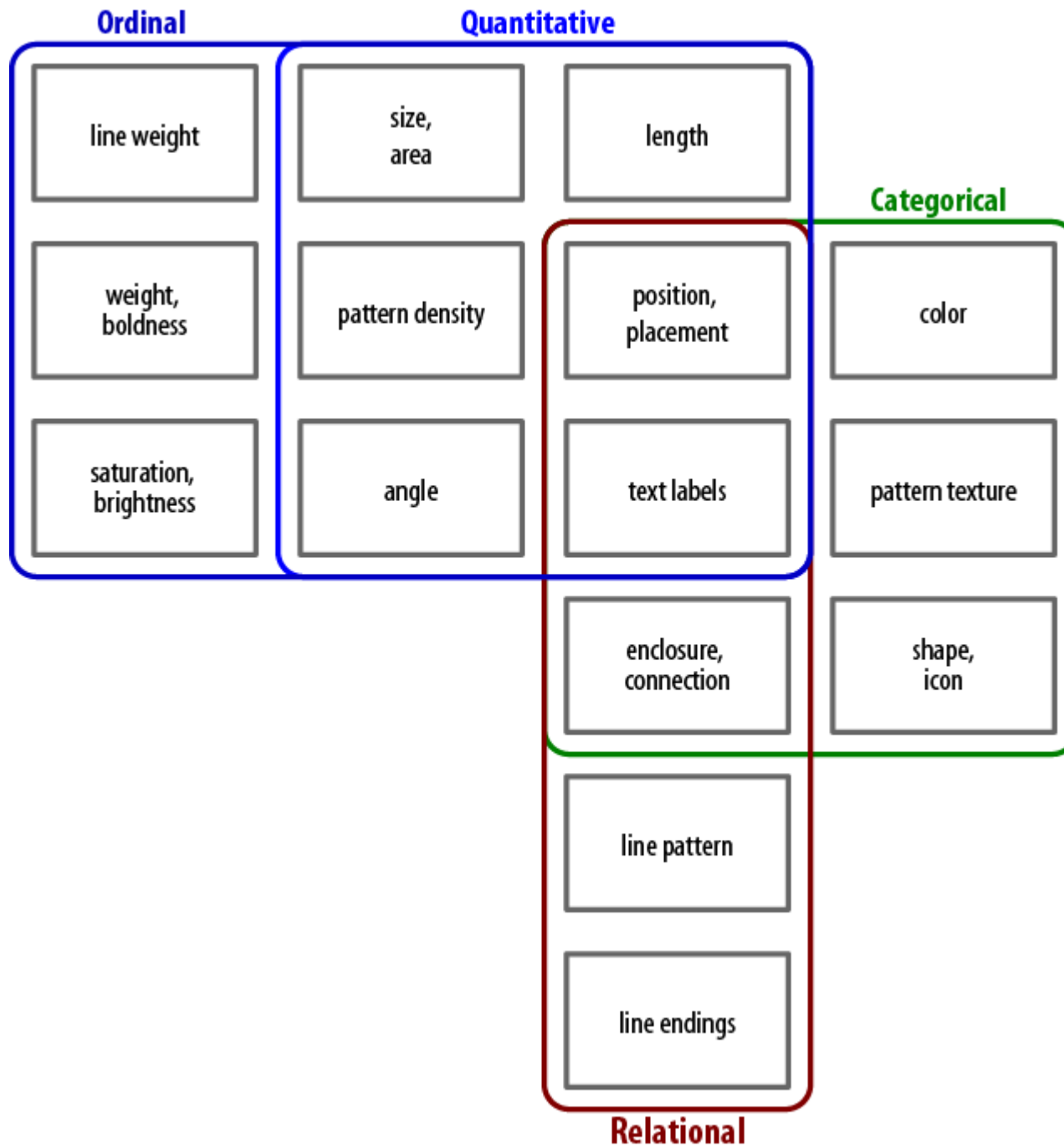
Some properties have intrinsic meaning but some don't:

- Density (Greyscale)
 - Darker -> More
- Size / Length / Area
 - Larger -> More
- Position
 - Leftmost -> first, Topmost -> first
- Hue
 - ??? no intrinsic meaning



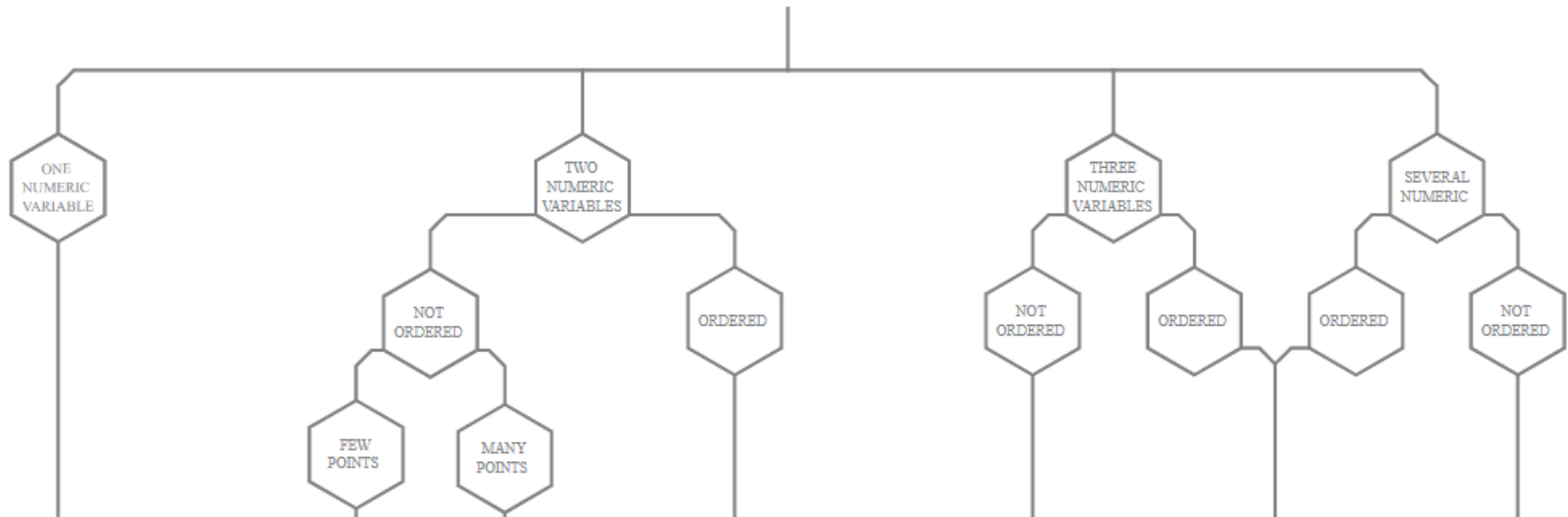
Additional visual channels (unranked): connection, containment/enclosure, crispness/resolution, flicker, line endings, line pattern, line weight, numerosity, text, texture, transparency, weight/boldness.

Example	Encoding	Ordered	Useful values	Quantitative	Ordinal	Categorical	Relational
	position, placement	yes	infinite	Good	Good	Good	Good
1, 2, 3; A, B, C	text labels	optional alpha or num	infinite	Good	Good	Good	Good
	length	yes	many	Good	Good		
	size, area	yes	many	Good	Good		
	angle	yes	medium	Good	Good		
	pattern density	yes	few	Good	Good		
	weight, boldness	yes	few		Good		
	saturation, brightness	yes	few		Good		
	color	no	few (<20)			Good	
	shape, icon	no	medium			Good	
	pattern texture	no	medium			Good	
	enclosure, connection	no	infinite			Good	Good
	line pattern	no	few				Good
	line endings	no	few				Good
	line weight	yes	few		Good		



From Data to Viz

What kind of data do you have? Pick the main type using the buttons below. Then let the decision tree guide you toward your graphic possibilities.



Appropriate Visual Encodings

Choosing Appropriate Visual Encodings

- Natural ordering
- Distinct values
- Defaults versus innovative formats
- Readers' context
- Compatibility with Reality
- Patterns and Consistency

Natural Ordering

Determined subconsciously by the mechanics of our visual system

Process is deeply embedded in our understanding and evaluates relative order independent of language, culture or convention

Depending on the specifics of the visual property, its natural ordering may be well suited to representing:

- Quantitative differences

- Ordinal differences

Examples of Natural Ordering

Position has a natural ordering

- shape does not

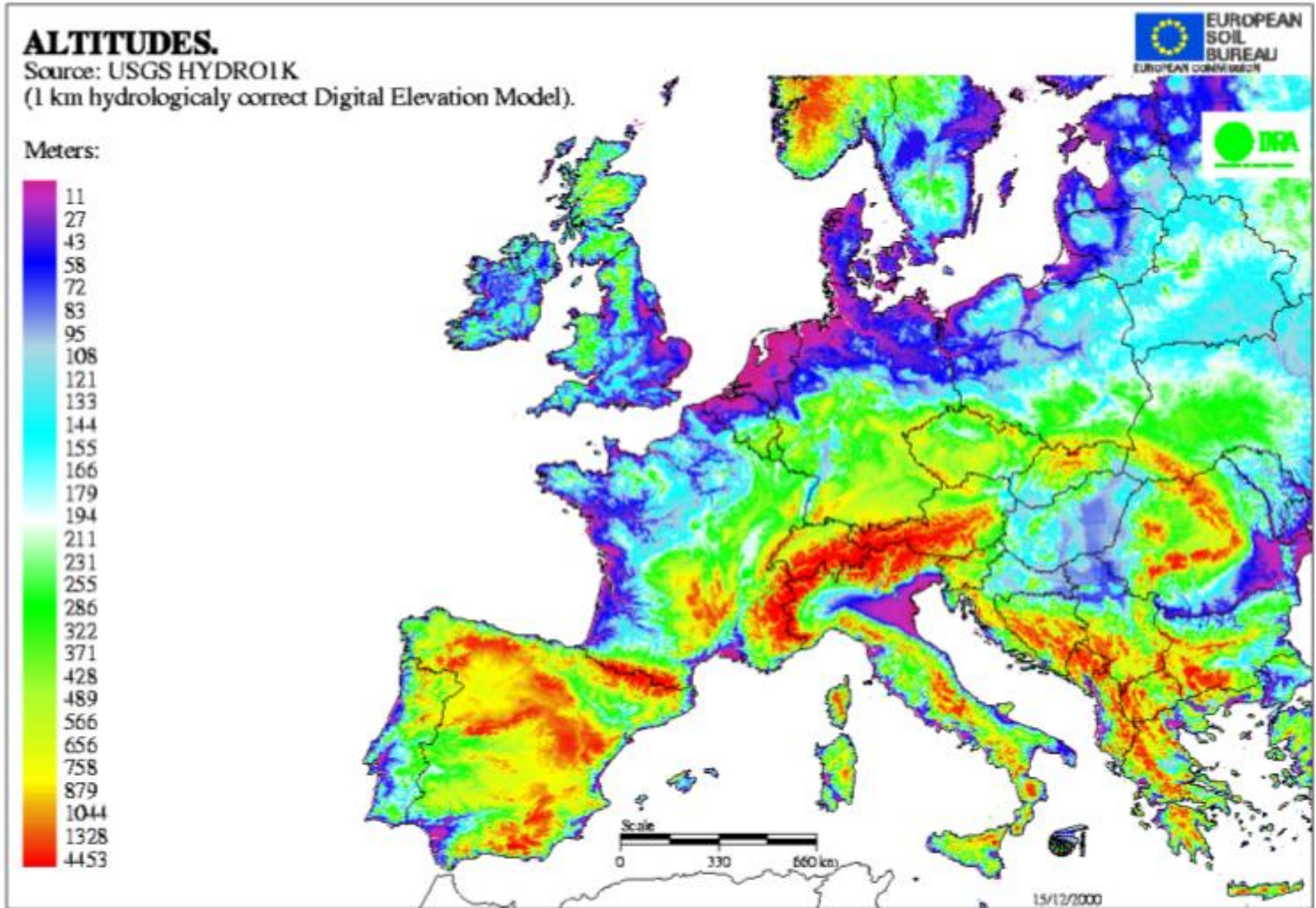
Length has a natural ordering

- texture does not (but pattern density does)

Line thickness or weight has a natural ordering

- line style (solid, dotted, dashed) does not

Colour Is Not Ordered



Colour Is Not Ordered

Colour (hue) is not naturally ordered in our brains

- brightness (lightness or luminance, sometimes called tint) is
- intensity (saturation) is

Strong social conventions about colour

Misuse of Colour

Misuse of colour to imply order is rampant

If tempted to use *ordered colour* (elevation, heat maps, etc.), consider varying brightness/Saturation

For example, elevation can be represented by increasing the darkness of the hue or with a divergent colour scale, rather than cycling through the rainbow

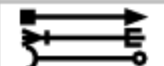


<https://www.istockphoto.com/photos/topographic-map-europe>

Distinct Values

- When choosing a visual property, select one that has a number of useful differentiable values and an ordering similar to that of your data
- There are a lot of colours in the world, but we cannot tell them apart if too similar
- We can more easily differentiate a large number of shapes, a huge number of positions, and an infinite number of numbers

Distinct Values

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	line weight	yes	few		Good		

Defaults vs Innovative Formats

- There are a lot of good default encodings and encoding conventions in the world, and with good reason...
 - Designing new encoding formats can cost you a lot of time and effort
 - Knowing the expected defaults for your industry, data type, or reader can save you a lot of work when it comes to figuring out how to best encode your data
- Sometimes default can put readers to sleep, but other times it helps them get to the heart of the matter...

Readers' Context

Audience and Culture

- The reader may show up with a mindset or way of viewing the world different from yours
- Is the reader from a culture that reads left-to-right, right-to-left, or top-to-bottom?
 - A person's habitual reading patterns will determine their default eye movements over a page, and the order in which they will encounter the various visual elements in your design
- May also pertain to geographic maps: many of us are used to seeing the globe split somewhere along the Pacific, with north oriented upward



<https://cartorica.com/shop/pacific-centered-world-map/>

McARTHUR'S UNIVERSAL CORRECTIVE MAP OF THE WORLD



Readers' Context

Convention and Colour Blindness

- Scientific universality about how we perceive and process colour but also significant cultural associations
 - Some colours may be lucky, some unlucky; some may carry positive or negative connotations; some may be associated with life events
 - Colours may also take on special significance when paired with certain shapes
- Many variations in the way different people perceive colour commonly called “colour blindness” but is more properly referred to as “colour vision deficiency” or dyschromatopsia
 - About 7-8% of men experience some kind of colour perception disorder. Women are much more rarely affected: about 0.4%-0.5%.

Compatibility With Reality

- Make your encodings of things and relationships as well aligned as possible with the reality (or your reader's reality) of those things and relationships
 - Some things are larger than others, have specific colours, well-known locations, and other identifying characteristics
- This alignment is called **compatibility**

Browser	Existing color	Better color
		
		
		
		
		

Compatibility With Reality

- Avoid countering conventions where possible to avoid creating **cognitive dissonance**
 - Clash of habitual interpretation with the underlying message you are sending
- Consider cultural conventions about spatial representations, such as what left and right mean politically



Patterns & Consistency

- As designers, we must be extremely deliberate about the patterns and pattern violations we create
- Three simple rules:
 - Be consistent in membership, ordering, and other encodings
 - Things that are the same should look the same
 - Things that are different should look different
- Maintaining consistency and intention when encoding will greatly enhance the accessibility and efficiency of your visualization

Patterns & Consistency

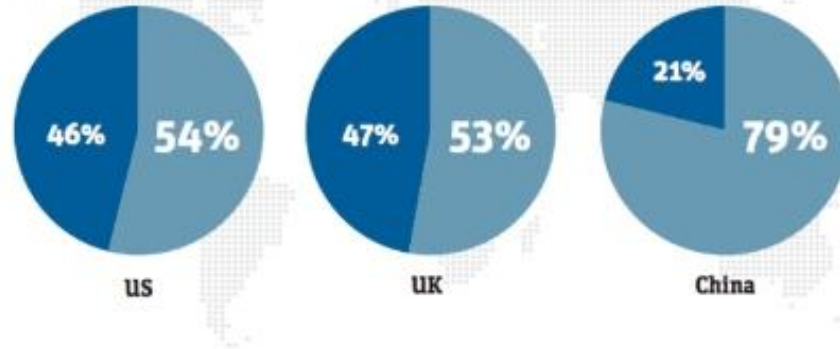
How many women are on your board of directors?

■ None ■ One or more



How many women are in your C-suite?

■ None ■ One or more



Do you have programs in place to increase the number of women in leadership positions?

■ Yes ■ No



View the complete Startup Outlook 2016 report at svb.com/leo

[Image: [Silicon Valley Bank](#)] —LD

Patterns & Consistency

China leads when it comes to female startup leaders with 79% of Chinese startups having one or more women in the C-Suite

How many women directors do you have on the board?

US



UK



CHINA



How many women are in your C-Suite?

US



UK

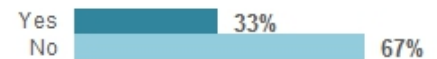


CHINA



Do you have programs in place to increase the number of women in leadership positions?

US



UK



CHINA



Source: Silicon Valley Bank, Startup outlook 2016

Thanks To

Cathy Ennis, John McAuley, Colman
McMahon and Brian Mac Namee
for an earlier versions of these lecture
notes